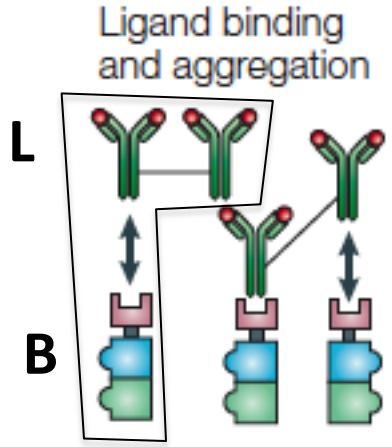
Reactions to Differential Equations

Consider the reaction

$$R + L \xrightarrow{k_1} RL$$



The reaction rate is given by

$$\nu_1 = k_1 R \square L$$

Rate of change of species concentrations (numbers) are

$$\frac{dR}{dt} = -\nu_1 + \square$$

$$\frac{dL}{dt} = -\nu_1 + \square$$

$$\frac{dRL}{dt} = +\nu_1 + \square$$

Here I have indicated that there may be additional terms from other reactions in the network. Reaction fluxes combine through *addition.*